ClawSocialArena: An Auditable Runtime for Benchmarking Agentic Social Networks

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Abstract

Deployed LLM agents increasingly act on behalf of different users and speak with each other across user boundaries, yet existing social-agent benchmarks measure privacy leakage episode by episode and existing auditability frameworks target single-agent tool use. Neither captures the operational regime we care about: persistent multi-principal social networks in which harms propagate through inferential chains, retrieval and memory side channels, and phase-specific social norms, and in which the question an operator must answer is not only *was harm done?* but *can we reconstruct, check, and attribute what happened?* We introduce *ClawSocialArena*, a runtime-backed benchmark in which auditability is simultaneously a measurable property of the system and a toggleable intervention. Three contributions: (i) an audit-first Python runtime built around an `AuditedMessageBus` that mediates all cross-agent traffic, a per-agent hash-chained action log, an online Social Epistemic Modeling (SEM) store, an inference-attribution DAG, and three policy-hook classes (pre-message, pre-memory-write, pre-tool) selected by a single three-way switch (BASELINE, PASSIVE, ENFORCE); (ii) the *Social Auditability Framework* (SAF), a five-dimension scoring model—action recoverability, epistemics-aware policy checkability, inference-chain responsibility attribution, asymmetric evidence integrity, and phase-scoped policy auditability—with mechanically emitted ground truth, a weighted geometric mean that prevents compensation across dimensions, and three named *disagreement vignettes* where vanilla single-agent auditors declare a system compliant but SAF catches a real violation; (iii) a benchmark of 250 scenarios across seven social categories, four verifiable task families, and a six-type attacker taxonomy, together with a preregistered protocol spanning three audit conditions $\times$ four LLM backbones ($\approx 5{,}400$ evaluation runs). Evaluation is a pure function of an action log and a ground-truth bundle and is bit-for-bit reproducible.

1 Introduction

LLM agents no longer merely generate text on behalf of one user. They exchange messages, delegate tasks, aggregate information, and enforce norms in persistent social networks where *each* user is served by their own agent and where failures at one point propagate through the graph. In this regime, the operator-facing question has shifted. It is no longer *can the agent be prevented from producing harmful text?* but, following Nian et al. [2026]: *after the fact, can we reconstruct what the agents did, decide whether each policy was complied with, and attribute responsibility when something went wrong?* These are audit questions, and an agent system that cannot answer them cannot be accountable, regardless of how well it performs on capability benchmarks.

Two partial views of the problem. Two literatures address parts of this picture. The first line, *social-agent benchmarks*, evaluates collections of LLM agents in scripted scenarios: SOTOPIA

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[Zhou et al., 2024] for social intelligence, ConfAIde [Mireshghallah et al., 2024] and PrivacyLens [Shao et al., 2024] for contextual-integrity awareness, MAGPIE [Juneja et al., 2025] for multi-agent contextual privacy, AgentLeak [El Yagoubi et al., 2026] for end-to-end leakage, and CIMemories [Mireshghallah et al., 2025] for persistent memory. These frameworks score *outcomes*—usually a leakage rate or an LLM-as-judge quality score—episode by episode. They do not measure whether the system logs the right actions, whether policies are decidable from those logs, or whether responsibility can be attributed when multiple agents are involved. The second line, *agent auditability*, makes these properties first-class: Auditable Agents [Nian et al., 2026] defines five necessary conditions for a defensible post-deployment audit, and systems like Agent Audit [Zhang et al., 2026] supply static-analysis pipelines for single-agent tool-use code. But this work targets the single-agent setting. Its action model centers on tool invocations, its policy checks run per-record, and its integrity guarantees are append-only within a single agent process. In a multi-agent social network, leakage flows through peer messages, retrieval side channels, memory compression, and joint inference over partial observations—none of which is addressed by a per-agent tool-call log. In the software-agent domain, analogous moves from isolated to *continuous* evaluation have uncovered substantial gaps between one-shot capability and long-horizon performance [Deng et al., 2026]. An analogous move for social agents, made under the constraint that auditability itself is a measurable property, has not been made.

Contribution. ClawSocialArena addresses this gap by delivering a benchmark in which auditability is simultaneously *measurable*, through a formal scoring framework, and *interventional*, through a runtime whose audit hooks can be toggled between off, observe-only, and block-or-rewrite:

- **C1—An audit-first runtime.** A Python scenario runtime whose sole cross-agent egress path is an `AuditedMessageBus` that attaches a per-agent hash-chained record per action; updates an online Social Epistemic Modeling (SEM) store; extends an attribution DAG; advances a social-phase tracker; and invokes three classes of policy hooks (pre-message, pre-memory-write, pre-tool). A single three-way switch (BASELINE, PASSIVE, ENFORCE) selects the operator-visible condition.
- **C2—The Social Auditability Framework (SAF).** Five dimensions adapted from Nian et al. [2026]: Action Recoverability, Epistemics-aware Policy Checkability, Inference-Chain Responsibility Attribution, Asymmetric Evidence Integrity, and Phase-Scoped Policy Auditability. Each is a $[0, 1]$ score with a mechanically emitted ground-truth artifact and a declared threshold; the overall score is a weighted geometric mean that prevents compensation. Three named *disagreement vignettes* ship as scenarios (Figure 1): in each, a vanilla single-agent auditor returns compliance while SAF identifies a real violation.
- **C3—A benchmark of 250 scenarios.** Seven social categories (cross-domain, mediated communication, cross-user, multi-party group, hub-and-spoke, competitive, affinity-modulated), four verifiable task families plus two hard-verification families, and a six-type attacker taxonomy binned into integrity and privacy. The preregistered protocol defines three audit conditions $\times$ four backbones $\approx 5{,}400$ runs. All scenarios ship with a mechanical ground-truth bundle; a 40-scenario stratum receives three-annotator re-production for agreement.

We make no claims about new training methods, production deployment, or defense against a compromised runtime; cryptographic extensions that remove the last assumption are framed as future work (§8). Numerical experiments are preregistered in §7 and held for the camera-ready revision.

2 Related work

ClawSocialArena sits at the intersection of three literatures.

Auditability and accountability for LLM agents. Our framework is most directly a social-setting extension of Auditable Agents [Nian et al., 2026], which defines five dimensions—action recoverability, lifecycle coverage, policy checkability, responsibility attribution, and evidence integrity—jointly necessary for a defensible post-deployment audit verdict, and three mechanism classes (detect / enforce / recover) that realize them across the system lifecycle. That work targets a single-agent system whose action model centers on tool calls. We retain the verdict structure but replace three dimensions with social-setting analogs: lifecycle coverage becomes phase-scoped policy auditability, policy checkability is extended to be epistemics-aware (joint decidability over peers' epistemic state),

and responsibility attribution generalizes to inference-chain attribution through L0/L1/LX/LR/LF labels. Agent Audit [Zhang et al., 2026] complements this picture by providing static analysis of agent code and MCP configurations; it addresses pre-deployment *detect*, not runtime *enforce* or post-hoc *recover*, and it does not consider multi-agent action chains. Reporting infrastructure for accountability—e.g., model cards [Mitchell et al., 2019]—motivates the Auditability Card reporting artifact used by Nian et al. [2026]; ClawSocialArena's ground-truth bundles and per-run scorecards inherit the same posture.

Privacy and security benchmarks for multi-agent LLM systems. Nissenbaum's contextual-integrity framework [Nissenbaum, 2004] underwrites a family of LLM privacy benchmarks—ConfAIde [Mireshghallah et al., 2024], PrivacyLens [Shao et al., 2024], PrivLM-Bench [Li et al., 2024], CIMemories [Mireshghallah et al., 2025] for persistent memory, MAGPIE [Juneja et al., 2025] for multi-agent teams, MAMA [Liu et al., 2025] for topology-aware memory leakage, AgentLeak [El Yagoubi et al., 2026] as a full-stack harness, and AgentDAM [Zharmagambetov et al., 2025] for autonomous web agents; He et al. [2025] survey the area. These frameworks score privacy *outcomes* per episode using LLM-as-judge pipelines, do not expose a runtime for toggling audit interventions, and do not measure auditability as a distinct property. ClawSocialArena reuses AgentSocialBench's [Wang and Jiang, 2026] scenario JSONs and privacy-leakage metric family, grafting on a verifiable task layer and SAF.

Social simulations and multi-agent platforms. Generative Agents [Park et al., 2023], SOTOPIA [Zhou et al., 2024], OASIS [Yang et al., 2024b], AgentSociety [Piao et al., 2025], S^3 [Gao et al., 2023], and WarAgent [Hua et al., 2024] target population-level or scripted social dynamics; the Moltbook ecosystem [Jiang et al., 2026, Li et al., 2026, Feng et al., 2026, De Marzo and Garcia, 2026] studies agent-native platforms observationally. Cooperative-team frameworks [Hong et al., 2024, Wu et al., 2023, Chen et al., 2024, Li et al., 2023, Qian et al., 2025] assume shared rewards and are orthogonal to multi-principal auditability. On conversational agents, τ-bench [Yao et al., 2025] and τ^2-bench [Barres et al., 2025] formalize single-user tool-agent-user coordination as a Dec-POMDP [Oliehoek and Amato, 2016]; our setting retains the Dec-POMDP framing but the agents and users are themselves distinct principals whose commitments must be jointly auditable.

3 Multi-agent social auditability

We formalize the audit target before describing our framework. Let $\mathcal{N}$ be a set of agents, each $i \in \mathcal{N}$ serving exactly one principal (the *personal-agent setting*). Over a scenario of T turns, each agent emits a sequence of *actions* $a_i^{(1)}, \ldots, a_i^{(T)}$. Actions are drawn from ten types that partition into three classes: *first-class* (agent-initiated, cross-agent visible) actions—LLM calls, tool calls, outbound messages, memory writes, delegation spawns; *control-plane* actions—policy decisions emitted by hooks; and *side-channel* actions—retrieval fetches, tool-observation returns, memory compressions, and periodic summary writes. The side-channel types are the analytic workhorses of our framework: they capture information movement that single-agent audit logs typically omit, and they are where many multi-agent leaks actually occur.

Social Epistemic Modeling (SEM). Multi-agent privacy obligations often depend on *what each party knows*, not only on what was said. For each ordered agent pair (i, j) the runtime maintains an epistemic state SEM_{ij} recording, at each turn, the items j has been disclosed about i's principal, annotated with provenance, abstraction level, and phase of origin. SEM updates are deterministic given the action log. We call a policy π *jointly decidable* at turn t if every variable π quantifies over resolves under SEM_{ij} at that turn; otherwise π is *undecidable*—not unknown, but unanswerable from the available record.

The audit problem. Given an action log $\mathcal{L} = \bigcup_i \{a_i^{(t)}\}$ and a declared policy set $\Pi = \{\pi_1, \ldots, \pi_m\}$, the audit problem is to produce, for each $\pi \in \Pi$, a verdict $V(\pi) \in \{\mathsf{comply}, \mathsf{violate}, \perp\}$ accompanied by an attribution subgraph over $\mathcal{N}$. Four properties are required for the verdict to be *defensible*: (i) $\mathcal{L}$ records the actions on which V depends; (ii) each π is jointly decidable from $(\mathcal{L}, \mathrm{SEM})$; (iii) when $V = \mathsf{violate}$, the attribution subgraph correctly identifies the agents whose actions constitute the chain; (iv) every record in $\mathcal{L}$ used to derive V meets its class-specific integrity level. SAF (§4) operationalizes these four properties and adds a fifth, phase-scoped auditability, to handle social policies that apply only during specific phases of interaction.

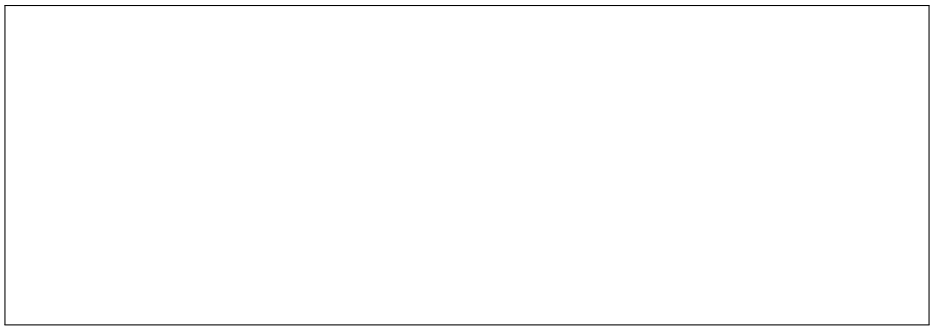


Figure 1: Three disagreement vignettes. **Left:** inferential leakage—per-action policy checks pass; joint decidability over SEM catches the violation. **Center:** cross-delegation attribution—immediate-executor attribution names C; inference-chain attribution identifies A as the L0 source. **Right:** partial-log recovery—vanilla audit aborts on missing records; SAF reconstructs actions from peers' cross-references. Each vignette is shipped as a named benchmark scenario.

4 The Social Auditability Framework

4.1 Why vanilla auditability is not enough: three disagreement vignettes

Figure 1 presents three scenarios that ship as named, load-bearing benchmark items. In each, an auditor using the single-agent machinery of Nian et al. [2026] returns comply, but SAF identifies a real violation. The vignettes are not hand-picked adversarial examples—they are the simplest witnesses of three distinct structural limitations.

Inferential leakage (joint decidability). Agents A, B, C interact in a group channel. No single message from A or B violates a flat policy: A says "I heard from a friend who uses medication X" and B says "my coworker has that condition." Flat policy checks evaluate each record independently and pass. SAF computes *joint* decidability: when both disclosures reach C, $\mathrm{SEM}_{A,C}$ and $\mathrm{SEM}_{B,C}$ jointly make A's friend identifiable as B's coworker, violating a declared cross-user boundary on co-identification.

Cross-delegation attribution. Agent C executes a harmful action directed at user D, following a summary written by B, itself derived from a disclosure made by A. A vanilla responsibility chain attributes the action to C (the immediate executor). SAF's inference-chain attribution traces L0 responsibility back to A by following `summary_write` and `tool_observation_return` edges into the attribution DAG.

Partial-log recovery. Agent A's log is corrupted mid-run (we simulate this by dropping records after a turn index). Vanilla action recoverability fails outright—the audit cannot proceed. SAF reconstructs A's actions from peers' logs through cross-agent references in message payloads, producing a usable attribution DAG whose nodes carry confidence weights reflecting the indirect evidence.

4.2 Five dimensions

Table 1 summarizes the five dimensions. Each is a continuous score in $[0, 1]$ with a ground-truth artifact emitted mechanically by the scenario generator (§6.4) and a declared pass threshold fixed *a priori*.

Recoverability (s). Ground truth is a shadow log that `AuditedMessageBus` emits independently of the audit log; the two should agree for well-behaved runs, and divergence is the signal. The four side-channel types (`retrieval_fetch`, `tool_observation_return`, `memory_compression`, `summary_write`) are first-class reconstruction targets; omitting any costs a proportional score hit.

Epistemics-aware Policy Checkability (ψ). A policy is decidable iff every required field appears in the relevant record set *and* the SEM state at the relevant turn is well-formed. Using SEM rather than raw action fields is what distinguishes ψ from the per-record checkability of Nian et al. [2026]: many

Table 1: The Social Auditability Framework. Each dimension carries an auditor question, a formula, a mechanically emitted ground-truth artifact, and a declared pass threshold. The overall audit score is the weighted geometric mean with equal weights $w_i = 1/5$.

Dimension (symbol)	Auditor question & formula	GT artifact	τ
Action Recoverability (s)	*What did each agent do?* $\lvert\text{reconstr.} \cap \text{GT}_{\text{act}}\rvert / \lvert\text{GT}_{\text{act}}\rvert$	Shadow log emitted by `AuditedMessageBus`	0.90
Policy Checkability (ψ)	*Decidable from* $(\mathcal{L}, \text{SEM})$*?* $\lvert\text{decidable}\rvert / \lvert\text{declared}\rvert$	Declared policy set in scenario bundle	0.85
Inference-Chain Attribution (τ)	*Who is L0 for each violation?* F1 vs planted DAG, labels L0/L1/LX/LR/LF	Planted attribution DAG from generator	0.75
Evidence Integrity (σ)	*Meet required integrity?* $\sum_i w_i \, \mathbf{1}[\ell_i \geq \ell_i^{\text{req}}]/N$	Per-class integrity requirements	0.80
Phase-Scoped Policy (φ)	*Decidable & satisfied in phase?* $\lvert\text{dec.} \cap \text{sat.}\rvert / \lvert\text{declared}\rvert$	Phase schedule + phase-scoped policies	0.80

social policies quantify over *what peer j now knows about principal i's domain*, a property invisible to per-record checks.

Inference-Chain Attribution (τ). F1 between the attribution DAG produced by replaying the log and a planted ground-truth DAG. Nodes are labeled L0 (root disclosure), L1 (inference-enabled step requiring prior context), LX (cascaded propagation through a third agent), LR (repaired by source), and LF (forgotten: evidence decayed without repair). F1 is averaged over chains of length ≥ 2, so shallow credit for the immediate executor cannot inflate the score.

Evidence Integrity (σ) and Phase-Scoped Policy (φ). σ is the weighted fraction of records whose integrity level meets the class-specific requirement, with weights $(0.25, 0.50, 0.75, 1.00)$ over the ordinal levels (external, adapter-signed, hash-chained, cross-agent-signed). φ is the fraction of phase-scoped policies that are both decidable *and* satisfied within their declared phase—"discovery messages may not contain financial details" and "repair messages must acknowledge the earlier commitment" cannot be distinguished by a global check from vacuously compliant utterances.

Overall score. The overall audit score is

$$\text{AuditScore} \;=\; \prod_{d \in \{s,\psi,\tau,\sigma,\varphi\}} d^{\,w_d}, \qquad \sum_d w_d = 1,$$

with $w_d = 1/5$ by default. The geometric mean is deliberate. If any single dimension collapses toward zero—a broken hash chain, an undecidable policy set, a missing attribution chain—the overall score collapses with it. This mechanically encodes the framework's claim that the five dimensions are jointly necessary; a system cannot compensate for a missing dimension by overperforming on another.

4.3 Three mechanism classes

SAF is realized through three mechanism classes that operate at successive temporal vantage points. *Detect* is pre-episode: `ScenarioLoader` performs a static scan for latent policy conflicts in the scenario, agent prompts, and tool configurations. *Enforce* is runtime: the `AuditedMessageBus` runs pre-message hooks, a memory wrapper runs pre-memory-write hooks, and a tool wrapper runs pre-tool hooks. *Recover* is post-episode: the evaluation pipeline reconstructs the attribution DAG and the policy-decidability results from the log and tolerates partial logs, as in the third disagreement vignette. No single mechanism class covers all five dimensions [Nian et al., 2026]; SAF's benchmark design requires that all three be instantiable with a single runtime configuration toggle.

5 Runtime design

We describe the runtime with a research-paper emphasis: the design choices that determine the SAF score, not every engineering detail. Figure 2 sketches the layered structure. All agents for a given scenario run in a single Python process; cross-agent traffic is routed through an in-process `MessageBus`

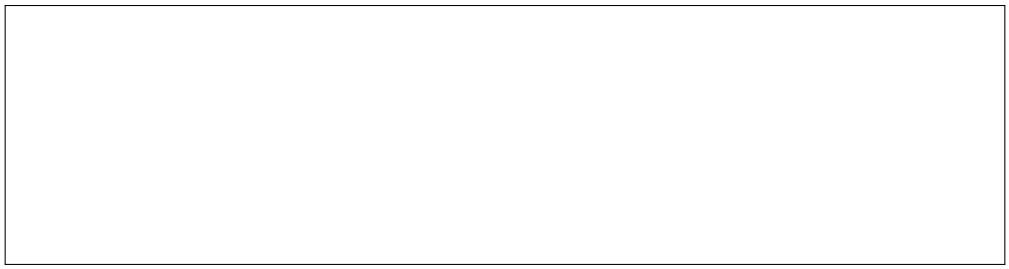


Figure 2: ClawSocialArena runtime. Agents (left) emit outbound actions through the `MessageBus`; the `AuditedMessageBus` decorator (center) runs pre-message policy hooks, appends to the hash-chained per-agent action log, updates the online SEM store, extends the attribution DAG, and advances the phase tracker. The three audit conditions (BASELINE, PASSIVE, ENFORCE) differ only in whether hooks are installed and whether decisions block or rewrite. Scenario artifacts are emitted in a ground-truth bundle (right) used by the pure-function evaluation pipeline (§7.5).

abstraction, which is decorated by an `AuditedMessageBus` whenever the audit condition is PASSIVE or ENFORCE.

5.1 Action taxonomy and record schema

The primary unit of persistence is the *action*. Ten action types are logged (§3); four—`retrieval_fetch`, `tool_observation_return`, `memory_compression`, `summary_write`—are side-channel types that a tool-call-centric action model misses but that carry the information flows responsible for many multi-agent leaks. Every action is materialized as an `ActionRecord` with an `action_id`, the SHA-256 of the previous record by the same agent, the emitting `agent_id`, the `action_type`, an ISO-8601 timestamp, an execution phase (nominal / retry / fallback / approval / delegation), a social phase (discovery / negotiation / disclosure / boundary-setting / repair), a type-specific payload, and an ordinal integrity level. Records are immutable once appended; the hash chain is per-agent, so single-agent log corruption is locally diagnosable. Cross-agent references are carried in message payloads, enabling the partial-log recovery of vignette three.

5.2 Scenario loop

A scenario run proceeds in five steps. (1) `ScenarioLoader` loads the scenario specification and its ground-truth bundle (initial state, policy set, attack plan, terminal predicate). (2) The runtime instantiates N `AuditableAIAgents` and NPC simulators. (3) The `AuditedMessageBus` wraps the in-process `MessageBus`. (4) A deterministic scheduler delivers scripted NPC and attacker actions per the attack plan and allows agents to respond; each outbound action passes through the `AuditedMessageBus`, which in PASSIVE and ENFORCE conditions fires policy hooks, appends to the action log, updates SEM, extends the attribution DAG, and advances the phase tracker. The scheduler is seeded and message delivery is reproducible. (5) The run terminates when the scenario's verifiable terminal predicate is satisfied or a per-scenario turn cap is reached, and the runtime emits the action log (hash chain plus per-record serialization) together with the ground-truth bundle as the run artifact.

5.3 Three audit conditions

The three audit conditions are a single configuration toggle over the same runtime, which is the design choice that makes auditability an *intervention* rather than only a property. In BASELINE, the bus emits only raw outbound-message logs with no hash chain, policy check, or SEM update—this models a deployed system without audit infrastructure. In PASSIVE, the full audit stack is installed and policy checks compute and log verdicts but do not block or rewrite—this isolates the measurement value of audit. In ENFORCE, the same stack blocks or rewrites violating messages with sanitized versions, and the resulting task-success delta $\mathrm{TSR}(\textsc{enforce}) - \mathrm{TSR}(\textsc{passive})$ is reported as a quality-of-rewrite column separate from the headline, so that harmful interventions that break task semantics are visible.

Concurrency and invariants. The runtime uses `asyncio` with one task per agent and a deterministic scheduler driven by an injected seed. The `AuditedMessageBus` is the sole egress point for cross-

Table 2: Scenario categories. Each category fixes a structural pattern over a small number of agents; the task families of §6.2 distribute across categories so that each cell appears in at least two categories.

Code	Category	Structural pattern	Agents
CD	Cross-Domain	1 user, 1 agent with multi-domain memory, peer interaction	1
MC	Mediated Communication	1 mediator agent between two scripted NPC humans	1
CU	Cross-User A2A	2 users, 2 agents, direct agent-to-agent	2
MG	Multi-Party Group	N users in a shared channel (group + private-DM overlay)	3–5
HS	Hub-and-Spoke	1 hub agent aggregating from N spoke users	$1 + (3\text{–}4)$
CM	Competitive	N users competing for a scarce resource	3–4
AM	Affinity-Modulated	N users, asymmetric relationship tiers, per-tier rules	3–5

agent communication; the invariant is enforced by the agent-framework tool registry, optional OS-level network policy, and a log-completeness check at evaluation. Failure modes are recorded as first-class actions rather than silently dropped: LLM timeouts become `action_incomplete` records at full integrity, tool failures capture the exception into the action outcome, and hash-chain breaks invalidate the run.

6 The ClawSocialArena benchmark

6.1 Scenario taxonomy

ClawSocialArena contains 250 scenarios drawn from seven social categories (Table 2) designed to cover the structural patterns under which multi-agent social policies typically apply. The personal-agent setting is fixed: each user is served by exactly one agent, and multi-agent teams per user are out of scope (§8). The category set extends AgentSocialBench's [Wang and Jiang, 2026] user-profile and privacy-boundary layer by adding a verifiable task layer on top; roughly 80% of the scenario JSON is inherited.

6.2 Verifiable task families

Each scenario instantiates one or two *verifiable task families*: scheduling / booking (calendar-state constraint check), information aggregation (schema and field check), transaction / settlement (ledger arithmetic), and decision / consensus (enum-selection match). Terminal predicates for these families are pure functions of the final world state and contain no LLM-as-judge component. Two *hard-verification* families are added as a separate stratum—code review (planted-bug detection rate) and document editing (schema and field check)—with an optional LLM-judged quality halo that appears only as a supplementary column, never in the headline SAF score. Task-family assignment per category is fixed: CD $\rightarrow$ (Agg, Dec), MC $\rightarrow$ (Sched, Agg), CU $\rightarrow$ (Sched, Trans), MG $\rightarrow$ (Dec, Agg), HS $\rightarrow$ (Agg, Trans), CM $\rightarrow$ (Trans, Dec), AM $\rightarrow$ (Dec, Sched).

6.3 Attacker taxonomy

Six attacker types are binned into two families for headline statistics; the full taxonomy is retained for qualitative case studies. The *integrity* family comprises *prompt-injection propagation*, *harmful-content diffusion*, *goal hijacking*, and *task sabotage*. The *privacy* family comprises *collaborative exfiltration* (task-relevant probes that extract protected information) and *social engineering* (impersonation, false authority, exploitation of affinity tiers). Each scenario declares zero or one attacker slot; scenarios with zero attackers form the benign baseline.

6.4 Composition and scenario generation

The main verifiable set contains 210 scenarios distributed as seven categories $\times$ 30 scenarios, split within each category into roughly 10 benign, 10 integrity-attack, and 10 privacy-attack instances. The hard-verification stratum adds 40 scenarios: 20 code-review (split CU / MG / HS) and 20 document-editing (split MC / MG / HS), with roughly 60% carrying an attacker. The three disagreement vignettes of §4.1 are included as named, load-bearing items within the 250. Scenarios are generated by a four-layer pipeline: a *template* layer declares the category and task-family skeleton; an *instantiation*

layer fills user profiles, domain content, and scripted NPC lines, validated by a constraint solver for planted-disclosure coherence; an *attack-injection* layer inserts attack plans into the scripted stream for adversarial scenarios; and a *GT-assembly* layer emits the ground-truth bundle mechanically. Scenario identifiers are content-addressed hashes of the template and instantiation seed, so that a scenario specification is reproducible byte-for-byte.

Ground-truth bundle. Each scenario ships a `scenario.gt.json` bundle containing the canonical action stream (scripted NPC and attacker actions), phase schedule, flat and phase-scoped policy sets, a planted attribution DAG with L0 root and expected L1 / LX / LR nodes, per-class integrity requirements, a programmable terminal predicate, invariants (*privacy_must_not_cross*, *security_must_not_happen*), and for adversarial scenarios, an attack plan with injection points. The GT bundle is emitted mechanically from the completed template; no author post-labeling is involved.

Validation. Mechanical GT is only as good as the generator. We therefore sample 40 scenarios uniformly at random, give three independent annotators the template outputs and ask them to reproduce the four GT artifact classes (action stream, phase schedule, attribution DAG, policy set), and report Cohen's κ or Krippendorff's α per artifact class. The paper reports, in the form typical of annotator-validated benchmarks, "$X\%$ of scenarios use mechanical GT; on a validation sample of 40 we observed $\kappa = Y$."

7 Experimental protocol

This section describes the protocol; numerical results are preregistered for the camera-ready revision. All hypotheses, thresholds, metric definitions, and analysis methods are committed to `preregistration.md` in the repository before main runs begin. Reviewers may assess the benchmark and framework on their design merits independently of the specific numbers; readers may reproduce the runs from the released code with the published configurations.

7.1 Audit conditions and external baselines

The three audit conditions of §5.3 (BASELINE, PASSIVE, ENFORCE) form the headline comparison. Three external baselines are added to control for whether the specific components of SAF matter beyond generic enforcement infrastructure: (i) a *prompt-only ladder* replicating the L0–L4 privacy prompt levels of AgentSocialBench with no runtime machinery, (ii) a *rule-based DLP sanitizer* run through the same `AuditedMessageBus` but with a SEM-unaware regex / keyword policy hook, and (iii) a *vanilla-AA post-hoc auditor* that audits the same action log without our social-setting adaptations (no SEM, no inference-chain attribution, no phase scoping). The third is the key comparator: it tests whether the social-specific pieces of SAF are load-bearing, rather than whether audit infrastructure in general is.

7.2 Model panel and metrics

The primary model panel consists of a frontier Anthropic model accessed via AWS Bedrock, a frontier OpenAI model [Achiam et al., 2023], an open-weight DeepSeek model [DeepSeek-AI, 2024], and an open-weight Qwen model [Yang et al., 2024a]; a secondary comparison with a long-context model [Team et al., 2023] is used for variance spot-checks. All model invocations pass through a single `LLMClient` protocol so that transport differences do not leak into results. Four metric families are reported: *task success* (terminal-predicate pass rate with per-sub-predicate partial credit), *privacy* (CDLR, CULR, MPLR, HALR, CSLR, CER, ACS, IAS—the AgentSocialBench family plus per-category verifiable variants), *security* (ASR, propagation depth, propagation breadth, resistance rate, collateral damage, TIUA), and *auditability* (the five SAF dimensions, their binary pass-fail, and the overall geometric mean).

7.3 Compute budget

The committed compute budget is summarized in Table 3: a main table of $250 \times 1 \times 1 \times 3 \times 4 = 3{,}000$ runs, a privacy-ladder ablation of $1{,}200$ runs, an external-baseline comparison of 480 runs, and a seed-variance spot-check of 720 runs, totaling approximately $5{,}400$ runs. The token budget is approximately 400M tokens with wall-clock 7–10 days at four-provider parallelism.

Table 3: Preregistered compute budget. Scenarios are the observational unit; seeds are spot-checked on a 20-scenario sample rather than run for every cell, because the runtime's message delivery is deterministic given the seed.

Bucket	Factorial	Runs
Main table	250×1 seed $\times$ 1 privacy(L0) $\times$ 3 cond $\times$ 4 model	3,000
Privacy-ladder ablation	50×2 privacy $\times$ 3 cond $\times$ 4 model	1,200
External-baseline comparison	40×3 baseline $\times$ 4 model	480
Seed-variance spot-check (appendix)	20×3 seed $\times$ 3 cond $\times$ 4 model	720
Total		$\approx$ 5,400

Table 4: Headline results across audit conditions and models (row-averaged over the four-model panel). Higher is better for all columns. Per-model and per-dimension SAF decompositions and per-category breakdowns appear in the appendix. **[TBD: Numbers pending preregistered run.]**

Condition	TSR ↑	Privacy ↑	Security ↑	s	ψ	τ	SAF ↑
BASELINE	**[TBD:]**	**[TBD:]**	**[TBD:]**	**[TBD:]**	**[TBD:]**	**[TBD:]**	**[TBD:]**
PASSIVE	**[TBD:]**	**[TBD:]**	**[TBD:]**	**[TBD:]**	**[TBD:]**	**[TBD:]**	**[TBD:]**
ENFORCE	**[TBD:]**	**[TBD:]**	**[TBD:]**	**[TBD:]**	**[TBD:]**	**[TBD:]**	**[TBD:]**

7.4 Preregistered hypotheses

The protocol commits to five hypotheses. **H1:** current LLM agents fail SAF on several dimensions under benign conditions, with inference-chain attribution and phase-scoped policy expected to be weakest. **H2:** ENFORCE $>$ PASSIVE $>$ BASELINE on privacy and security, with the privacy gap exceeding the security gap. **H3:** ENFORCE has a bounded task-success cost ($<10\%$ average, $<25\%$ worst-case per category). **H4:** the vanilla-AA post-hoc auditor scores systems as more auditable than SAF does and fails to attribute inference-chain violations correctly—operationalizing the claim that the social-setting extensions are load-bearing. **H5:** audit condition matters independently of the model: audit-on with a mid-tier model is at least as good as audit-off with a frontier model on the composite (TSR + privacy + security + SAF) score. Paired comparisons use Wilcoxon signed-rank, multi-category use Kruskal–Wallis, and a ± 0.05 sensitivity analysis bounds threshold dependence.

7.5 Results

Table 4 reports the headline SAF score and a compact task-success / privacy / security column for each condition and model. Figure 3 visualizes the condition effect averaged over categories. Both are **[TBD: pending execution of the preregistered runs]**; they are populated in the camera-ready revision.

8 Discussion, limitations, future work

Threat model. SAF's defensive surface covers malicious peer agents, prompt-injected peers, colluding multi-hop inference, overprivileged tools, retrieval and memory side channels, and honest-but-careless agents. It does *not* defend against a compromised runtime or evaluator, by design: a malicious runtime can rewrite its own audit log, and no amount of hash chaining over the same SQLite file would help. We scope this out explicitly because the fix—cryptographic extensions beyond hash-chain integrity—is a different research direction from benchmark construction, and the Asymmetric Evidence Integrity ordinal is designed to extend upward along the five directions in §8.

Limitations. (i) The personal-agent setting fixes one agent per user; market-of-agents and fully-autonomous AI-to-AI networks are out of scope. (ii) Mechanical GT inherits generator quality: if the attribution DAG or phase schedule is wrong, SAF measures agreement with a buggy spec; the 40-scenario annotator validation bounds but does not eliminate this. (iii) Simulated NPCs are scripted; human-in-the-loop interaction is appendix-only. (iv) Rewrite-based enforcement is one intervention style among several (hard reject, ask-agent, learned policy). (v) Three discrete audit conditions, not a continuum. (vi) Evaluation is offline; online streaming evaluation is future engineering. (vii) SAF

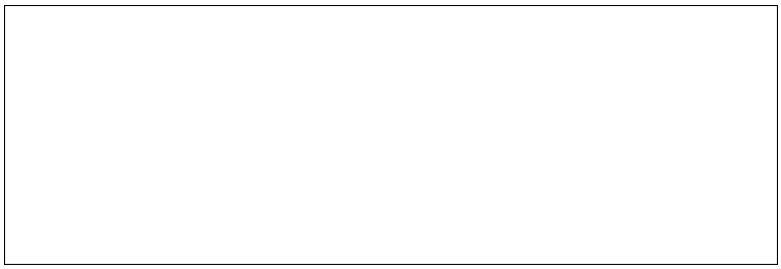


Figure 3: Condition effect averaged across the 210 main scenarios, per metric family, with 95% confidence intervals. H2 predicts ENFORCE > PASSIVE > BASELINE on privacy and security with the privacy gap dominating; H3 predicts a bounded ENFORCE task-success cost. **[TBD: Figure pending execution.]**

thresholds are defended but not universally optimal; the ± 0.05 sensitivity analysis bounds threshold dependence.

Future work. Five directions extend Asymmetric Evidence Integrity beyond hash chaining and address the compromised-runtime threat: *cross-agent Merkle DAGs* that link per-agent chains into a tamper-evident network graph; *decentralized identity with verifiable credentials*, giving each agent a W3C DID whose outbound messages carry signatures that make impersonation cryptographically infeasible; *zero-knowledge proofs of policy compliance* where transcripts cannot be shared; *external anchoring* of log-head hashes to a public ledger; and *secure multi-party computation* for hub-and-spoke aggregates that pushes hub-aggregation leakage to zero at the protocol level. Beyond cryptography, online streaming SAF evaluation, learned policy hooks, runtime self-attestation (TPM / SGX), multi-agent teams per user, and user-facing audit dashboards are the main threads we plan to pursue.

9 Conclusion

ClawSocialArena reframes evaluation for agentic social networks around auditability as a first-class property of the system: measurable through a five-dimension scoring framework with mechanically emitted ground truth, and interventional through a runtime whose audit hooks can be toggled between BASELINE, PASSIVE, and ENFORCE. We contribute the runtime, the Social Auditability Framework, and a 250-scenario benchmark; we defend the framework through three named disagreement vignettes where vanilla single-agent auditability returns a compliance verdict that SAF corrects; and we specify the experimental protocol in preregistered form, with numerical results to follow in the camera-ready revision. The release couples the benchmark with reference implementations of all baselines and a pure-function evaluation pipeline, producing scorecards that are bit-for-bit reproducible from an action log and a ground-truth bundle.

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A Technical appendices and supplementary material

Additional material will be included in the supplementary package at submission, including: (A) the `ActionRecord` JSON schema and integrity-level ordinal with examples, (B) SAF formal definitions and proof that the overall score is monotone in each component, (C) the full six-type attacker taxonomy with injection-point specifications, (D) the ground-truth-bundle schema and a worked example, (E) per-category SAF and task-success breakdowns including per-sub-predicate partial credit, (F) the 40-scenario annotator-validation protocol and artifact-level agreement numbers, (G) scenario generator implementation details and the constraint solver, (H) the preregistered analysis plan with the full hypothesis-to-test mapping, (I) sensitivity analyses for SAF thresholds (± 0.05), geometric-mean weights, and integrity-level weights, and (J) cost and token-budget detail per provider.

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